🧬 GENETIC ENGINEERING & BIOTECHNOLOGY — FULL MIND MAP

Biology Ch. 10 · Genetics & Biotechnology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Recombinant DNA, CRISPR, RNAi & mRNA
2. Mitochondrial Therapy, Cloning & IVF
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5. GM Mustard, Crop Engineering & Biopesticides
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9. Transgenics, Mutation, Pleiotropy & Gene Therapy

1. Recombinant DNA, CRISPR, RNAi & mRNA

- Microsatellite DNA = short, repeated (usually non-coding) DNA segments. Source inhe polymorphic markers ke roop mein inheritance/evolutionary relationships, DNA profiling, paternity/forensics, linkage analysis aur population genetics se link karta hai. *[Q1]*
- Source Q2 ke according different species ke DNA segments jodkar functional chromosomes banana NOT correct development hai; artificial functional DNA pieces, cell-free DNA replication aur lab petri-dish cell division ko possible statements maana gaya hai. *[Q2]*
- Recombinant DNA technology se bana pehla human hormone Insulin tha; source Humulin ko 1982 FDA approval se link karta hai. *[Q3]*
- Recombinant-vector vaccines mein genetic engineering use hoti hai aur live attenuated bacteria/viruses vectors ke roop mein desired antigen-gene carry kar sakte hain. *[Q4]*
- rDNA technology genes ko different plant species ke beech, animals se plants mein aur microorganisms se higher organisms mein transfer kar sakti hai. *[Q5]*
- Source ka rDNA sequence: genetic material identify/isolate → DNA fragment → desired fragment isolate → gene amplify → vector se ligate → recombinant DNA host mein transfer → foreign-gene product obtain → downstream processing. *[Q6]*
- Restriction endonucleases DNA ko specific sites par cut karne wali molecular scissors hain; DNA ligases fragments ko join karne wale molecular stitchers hain. *[Q7, Q8]*
- Cas9 (CRISPR-associated protein 9) targeted gene editing ki molecular scissors hai—DNA cut karke genome alter kar sakti hai. *[Q9]*
- RNA interference (RNAi) targeted mRNA ko neutralize karke gene expression/translation suppress karta hai. Source uses: gene-silencing therapy, cancer therapy aur viral-resistant crops; hormone-replacement therapy nahi. *[Q10]*
- mRNA ka expanded form Messenger Ribonucleic Acid hai; ye DNA ka genetic code nucleus se ribosome tak le jaata hai jahan proteins bante hain. *[Q11]*

2. Mitochondrial Therapy, Cloning & IVF

- Pronuclear Transfer / Mitochondrial Replacement Therapy ka purpose offspring mein mitochondrial diseases prevent karna hai; source ise 'three-parent baby' reproductive technology se link karta hai. *[Q12, Q13]*
- Dolly first mammal thi jo adult somatic cell se successfully clone hui. Source: Roslin Institute, Scotland; birth 5 July 1996; IVF se nahi bani; 2003 mein lung disease se death. *[Q14, Q15, Q16, Q17, Q18]*
- NDRI Karnal cloning: source Samrupa ko first cloned buffalo calf batata hai; second cloned calf Garima tha. Garima II bhi cloned buffalo hai; March 2023 ki 'Ganga' India's first cloned cow source ke according Gir breed ki thi. *[Q19, Q20, Q21]*

- National Taiwan University group ne 2006 mein green fluorescent protein introduce karke first transgenic glowing green pigs produce kiye. *[Q22]*
- Injaz duniya ki first female cloned camel thi, Camel Reproduction Centre, Dubai (UAE) mein developed. *[Q23]*
- Clone ki genetic information parent organism ke identical hoti hai aur clone asexually produce hota hai. *[Q24]*
- Test-tube baby/IVF mein fertilization mother's body ke bahar hota hai; fertilized embryo baad mein uterus mein transfer hota hai. *[Q25]*
- Source ke according genetic engineering mein disease resistance, growth promotion aur animal cloning kuch success ke saath achieve hue; human cloning ko fruitful result nahi mila/controversial-banned bataya gaya hai. *[Q26]*

> ⚠ **EXAM TRAP — Mitochondrial Inheritance — Source vs Key**
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> ▸ Q13 source marked answer (A) = statement 1 only. Explanation mein mtDNA ko usually maternal bataya gaya hai, 2018 research ke biparental exceptions mention hain, aur source note karta hai ki UPSC official key ne option (C) diya tha. Is mismatch ko source ke jaisa hi preserve kiya gaya hai. *[Q13]*

3. Genome Mapping, Stem Cells & Hybridoma

- Genetics scientist-matching source: bacterial transduction/conjugation → Lederberg; sex-linked inheritance → Morgan; E. coli DNA polymerase → Arthur Kornberg; complete genetic code → Hargobind Khorana. *[Q27]*
- Chromosome par genes/DNA sequences ki relative positions jaana livestock pedigree aur disease-resistant animal breeds develop karne mein useful hai; source ke according isse ALL human diseases ke causes nahi samjhe ja sakte. *[Q28]*
- 1996 mein budding yeast Saccharomyces cerevisiae ka genome completely sequence hua—source ise first completely sequenced eukaryotic genome se link karta hai. Human Genome Project = human genes aur sequences ki identification/mapping. *[Q29, Q30]*
- Stem cells self-renew/divide kar sakti hain aur many specialized cell types mein differentiate ho sakti hain; source inhe serious diseases ke research focus se link karta hai. *[Q31]*
- Stem cells mammals-only nahi; source ke according new-drug screening aur medical therapies mein use ho sakti hain. *[Q32]*
- Human stem-cell sources embryo tak limited nahi—fetus, umbilical cord, placenta, infant, child/adult aur organs/tissues se bhi mil sakti hain. Source Q33 mein in-vitro self-renewal virtually forever aur Indian cell lines ko correct maanta hai; embryonic stem-cell derivation ko blastocyst destruction se link karta hai. *[Q33, Q34]*
- Bone-marrow-derived stem cells ko source embryonic stem cells ka bioethically non-controversial alternative maanta hai. *[Q35]*
- Source CRISPR-Cas9 ke context mein germ-cell genetic changes, early-embryo genome editing aur human induced pluripotent stem cells ko pig embryo mein inject karne—teeno statements ko correct maanta hai. *[Q36]*
- Somatic Cell Nuclear Transfer (SCNT) reproductive cloning of animals mein use ho sakti hai; source therapeutic cloning / regenerative medicine aur embryonic stem-cell research bhi mention karta hai. *[Q37]*
- Hybridoma technology monoclonal antibodies ke commercial production ke liye hai. Technique Georges Köhler aur César Milstein ne 1975 mein develop ki; source 1984 Nobel se link karta hai. *[Q38, Q39]*

4. Gene Bank, Patents & Bt Crops

- Plant Field Gene Bank, Banthara ka source-purpose: endangered plants ko secure karna, biological-diversity piracy rokna aur economically important plants ko recognise karna. *[Q40]*

- Source ke Indian Patents Act discussion mein plants/seeds/varieties aur essentially biological processes patent-eligible nahi; plant varieties patent nahi ho sakti. Q41 explanation IPAB ke existence ko bhi state karta hai. *[Q41]*
- Bt cotton mein insect-resistance gene bacterium Bacillus thuringiensis se aata hai; Bt toxins insects ke liye harmful hain. Source assertion-reason mein Bt-gene insertion aur bacterial origin dono true + reason explains assertion. *[Q42, Q43, Q44, Q45]*
- Bollgard I & Bollgard II GM crop technologies Bt cotton se related hain; source Bollgard I = single-gene commercialisation 2002 aur Bollgard II = double-gene technology 2006 batata hai. *[Q46]*
- Bt brinjal genetically modified/transgenic brinjal hai; Cry1Ac gene soil bacterium Bacillus thuringiensis se insert karke insect/pest resistance (especially shoot/fruit borer) ke liye develop kiya gaya. *[Q47, Q48, Q49]*
- Bt brinjal opposition ke genuine concerns source mein health-impact aur biodiversity-impact (statements 3,4) hain; soil fungus gene aur terminator-seed claim ko incorrect maana gaya. *[Q50]*

5. GM Mustard, Crop Engineering & Biopesticides

- GM mustard = DMH-11 (Brassica juncea), source ke according Prof. Deepak Pental/University of Delhi team ne develop ki. Bacillus amyloliquefaciens genes cross-pollination/hybridization allow karte hain; broad pest resistance nahi; IARI+PAU joint-development statement incorrect. *[Q51]*
- Genetically engineered plants source mein drought tolerance, enhanced nutritive value aur longer shelf-life ke liye banaye gaye; spacecraft mein grow/photosynthesis wali statement accepted nahi. *[Q52]*
- Agriculture mein genome sequencing disease/drought-resistance markers identify karne, new crop varieties ka development time reduce karne aur host-pathogen relationship decipher karne mein useful hai. *[Q53]*
- Flavr-Savr tomato source ke according first commercially grown genetically engineered food tha; fruits longer firm rehte aur vine-ripening ke baad transport ho sakte; full vine-ripened flavour milta hai—A & B correct, C incorrect. *[Q54]*
- Transgenic crops create karne mein source cytoplasmic male sterility aur gene silencing ko use karta hai; budding/grafting vegetative propagation hai, genetic modification nahi. *[Q55]*
- Biopesticides mein bacteria, fungi aur flowering plants ki kuch species use ho sakti hain. Bacillus thuringiensis (Bt) microbial insecticide / biological insecticide ka classic example hai. *[Q56, Q57, Q58]*
- Terminator Seed Technology next generation ke liye non-viable/sterile seeds produce karti hai; source ise farmers ko har season seeds repurchase karne ki dependency se link karta hai. *[Q59, Q60, Q61]*

⚠ EXAM TRAP — Terminator Technology — Q61 Pattern

▸ Q61 mein defining feature sterile next-generation seed hai, lekin source explicitly note karta hai ki UPPSC ne 'All of the above' accept kiya because option-set GM plants ko broadly encompass karta tha. *[Q61]*

6. Golden Rice & Agricultural Biotechnology

- Golden Rice transgenic rice hai jo edible parts mein beta-carotene biosynthesize karta hai; beta-carotene body mein Vitamin A mein convert hota hai. Source isliye ise Vitamin-A deficiency/blindness combat karne se link karta hai. *[Q62, Q63, Q64, Q65, Q66, Q67, Q68, Q69]*
- Source Golden Rice ko Prof. Ingo Potrykus aur Dr. Peter Beyer se link karta hai; genome mein two daffodil genes + one Erwinia uredovora gene mention hain. Q70 ka marked answer Daffodil hai. *[Q70]*
- Incorrect match in source: Ozone layer → Troposphere, kyunki ozone layer Stratosphere mein hoti hai. Other matches: Renneting → Cheese; Plasmid → Genetic Engineering; Golden Rice → Vitamin A. *[Q71]*
- 'Super rice' source ke according G.S. (Gurdev Singh) Khush ne develop kiya; source IRRI, Philippines se link karta hai. *[Q72]*

7. Prenatal Diagnostics, Bioinformatics & Biotech Tools

- Amniocentesis fetal sex determine karne ke saath prenatal chromosomal abnormalities aur fetal infections diagnose karne mein use hoti hai; amniotic fluid se fetal DNA examine hota hai. *[Q73]*
- Transcriptome = organism/cell ke genes se expressed full range/sum-total of mRNA molecules; genome roughly fixed hota hai par transcriptome condition ke saath change kar sakta hai. *[Q74]*
- Arthur Kornberg ne DNA in vitro synthesize kiya; source unhe biological DNA synthesis mechanism discovery aur 1959 Nobel se link karta hai. *[Q75]*
- Biofilms medical implants aur food/food-processing surfaces par form ho sakte hain aur antibiotic resistance/tolerance show kar sakte hain—source teeno statements correct maanta hai. *[Q76]*
- Biochip biological environment ke liye designed microchip hai; source ke according DNA, RNA aur Protein chips/contents relevant hain. *[Q77]*
- Genico Technology ko source genetic diseases ki pre-information, especially prenatal fetal-disease investigation, se define karta hai. *[Q78]*
- Aerial metagenomics = air/aerosols se genetic material (DNA/RNA) collect karke airborne microbial communities ki genetic diversity analyse karna. *[Q79]*
- Biotechnology source-definition: living organisms, unse obtained substances ya biological processes ka industrial use; example yeast-based liquor production. *[Q80]*
- Recirculating Aquaculture System ke biofilters uneaten feed/waste treatment karte aur ammonia → nitrite → nitrate convert karte hain; phosphorus ko nutrient ke roop mein increase nahi karte. *[Q81]*
- Source 'recently evolved in genetic engineering' ka answer Gene Mapping deta hai; gene mapping chromosome par gene location aur relative distances determine karta hai. *[Q82]*
- Plasmid genetic engineering mein vector/carrier ke roop mein use hota hai. Useful bacteria: Escherichia coli + Agrobacterium (especially Ti plasmid in plant transformation). Bacteriophages bacteria ko infect karte aur genetic-engineering vectors ke roop mein use ho sakte hain. *[Q83, Q84, Q85]*
- Human growth-hormone gene se unusually large mouse produce karne wali technique source ke according Genetic Engineering hai. *[Q86]*

8. DNA Fingerprinting & Biometrics

- Criminal DNA testing ke samples source mein blood cells, bone cells, hair strands aur saliva—sab ho sakte hain. *[Q87]*
- DNA fingerprinting paternity aur criminals ki identity establish karne ka powerful tool hai; hairs, saliva aur dried semen jaise trace evidence DNA analysis ke liye adequate ho sakte hain. *[Q88]*
- DNA fingerprinting ka basis DNA polymorphism hai—individuals ke inherited variable DNA regions. *[Q89]*
- Child paternity prove/establish karne ke liye source DNA fingerprinting ko correct technique maanta hai. *[Q90, Q91]*
- Human identity establish karne ke biotechnological technique ke roop mein source DNA fingerprinting deta hai; applications: forensic cases, paternity disputes aur endangered living beings conservation. *[Q92, Q93]*
- Source ke according DNA fingerprinting se solve hua first crime England mein 1983 case se linked hai (explanation 1983 aur 1986 ke two murders mention karta hai). *[Q94]*
- Biometric identification mein fingerprint ke alawa iris scanning, retinal scanning aur voice recognition—teeno use ho sakte hain. *[Q95]*
- Multi-coloured/non-porous surfaces par fingerprints develop karne ke liye source Fluorescent powder ka answer deta hai. *[Q96]*

9. Transgenics, Mutation, Pleiotropy & Gene Therapy

- Source ke according transgenics se biodegradable plastic, edible vaccines aur transgenic crops achieve ho sakte hain; cloned animals ka production transgenics se achieve nahi maana gaya. *[Q97]*
- Gene ke base sequence mein permanent change = Mutation. *[Q98]*
- Ek gene jab do ya zyada different phenotypic characters ko simultaneously influence kare, phenomenon Pleiotropy kehlata hai. *[Q99]*
- Gene therapy defective genes ki functioning correct karne ke liye healthy gene copy introduce/replace kar sakti hai, faulty gene ko inactivate ('knock out') kar sakti hai, ya new/modified therapeutic gene introduce kar sakti hai. *[Q100]*

> ⚠ **EXAM TRAP — Gene Therapy — Multiple Correct**
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> ▸ Q100 source explicitly 'Multiple Correct Answers (Ans. *)' mark karta hai. Explanation ke according faulty gene ko healthy copy se replace karna, defective gene ko inactivate karna, ya new/modified gene introduce karna—multiple valid mechanisms hain. *[Q100]*